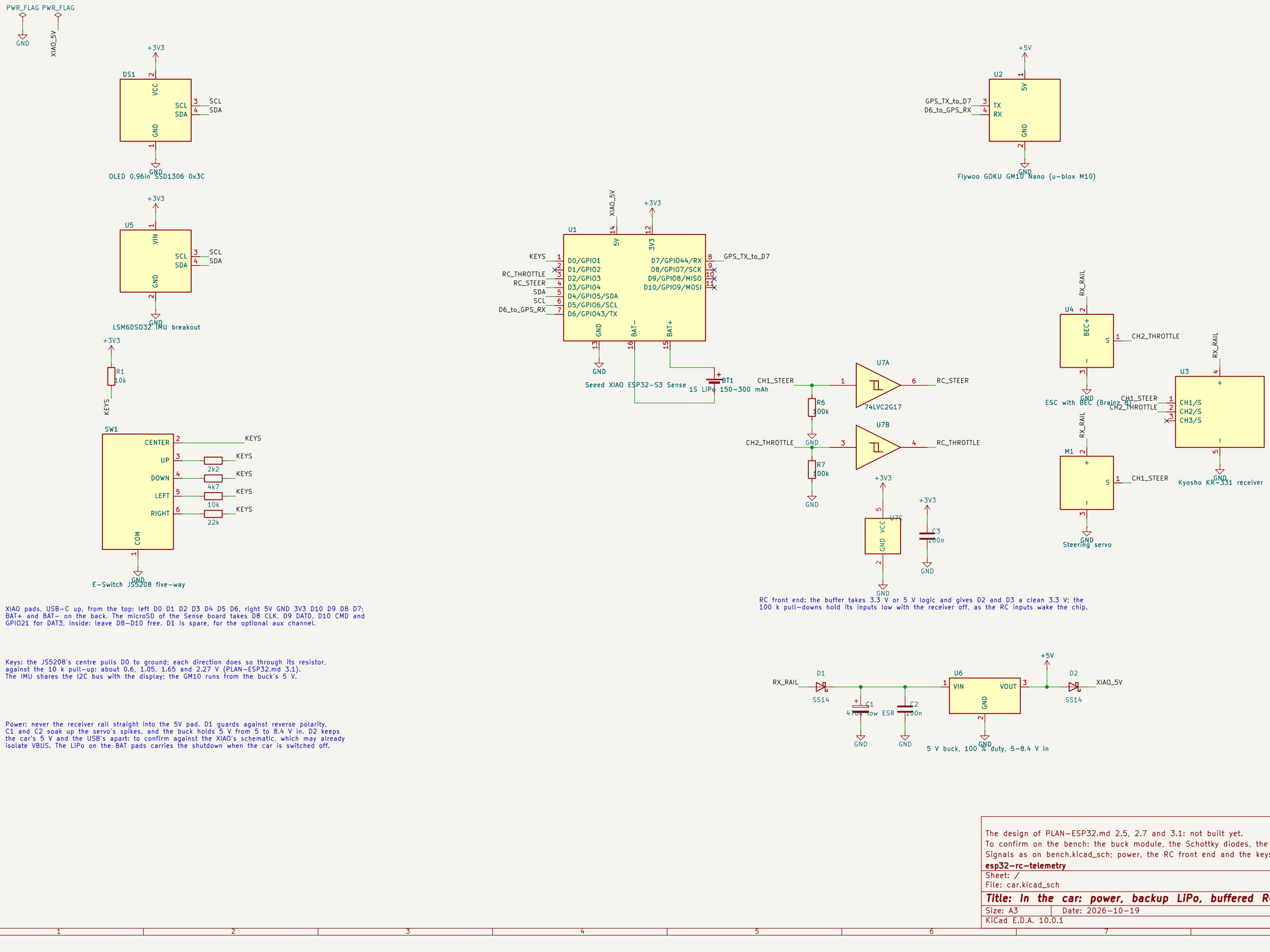


XIAO pads, USB-C up, from the top: left D0 D1 D2 D3 D4 D5 D6, right 5V GND 3V3 D10 D9 D8 D7; BAT+ and BAT- on the back. The microSD of the Sense board takes D8 CLK, D9 DAT0, D10 CMD and GPIO21 for DAT3, inside: leave D8-D10 free. D1 is spare, for the optional aux channel.

Keys: the JS5208's centre pulls D0 to ground; each direction does so through its resistor, against the 10 k pull-up: about 0.6, 1.05, 1.65 and 2.27 V (PLAN-ESP32.md 3.1).
The IMU shares the I2C bus with the display; the GM10 runs from the buck's 5 V.

Power: never the receiver rail straight into the 5V pad. D1 guards against reverse polarity, C1 and C2 soak up the servo's spikes, and the buck holds 5 V from 5 to 8.4 V in. D2 keeps the car's 5 V and the USB's apart: to confirm against the XIAO's schematic, which may already isolate VBUS. The LiPo on the BAT pads carries the shutdown when the car is switched off.

The design of PLAN-ESP32.md 2.5, 2.7 and 3.1: not built yet.
To confirm on the bench: the buck module, the Schottky diodes, the IMU breakout.
Signals as on bench.kicad_sch; power, the RC front end and the keys change.

esp32-rc-telemetry

Sheet: /
File: car.kicad_sch

Title: In the car: power, backup LiPo, buffered RC inputs, JS5208, IMU

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